NUMBER THEORY LADDER 5 OF 9

The Chinese remainder theorem

Residues modulo coprime numbers combine, so a number modulo 15 is a pair of numbers modulo 3 and modulo 5.

1 / THE RULE AND THE PICTURE

Each pair of remainders modulo 3 and 5 occurs for exactly one number below 15

$n \bmod 3$ \ $n \bmod 5$	0	1	2	3	4
0	**0**	**6**	**12**	**3**	**9**
1	**10**	**1**	**7**	**13**	**4**
2	**5**	**11**	**2**	**8**	**14**

the cell for $n \equiv 2 \pmod 3$ and $n \equiv 3 \pmod 5$ holds 8

If $\mathrm{HCF}(m, n) = 1$ the map $x \mapsto (x \bmod m,\ x \bmod n)$ is a bijection from the mn residues onto the mn pairs:

$$\boxed{\mathbb{Z}/mn \ \cong\ \mathbb{Z}/m \times \mathbb{Z}/n}$$

Every cell of the table is filled once, so the system $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$ has exactly one solution modulo 15, namely $x \equiv 8$.

2 / SUNZI'S PUZZLE / THREE MODULI

Find x with remainders $2, 3, 2$ on division by $3, 5, 7$

With $M = 105$ take $M_i = M/m_i = 35, 21, 15$ and the inverses $35^{-1} \equiv 2 \pmod 3$, $21^{-1} \equiv 1 \pmod 5$, $15^{-1} \equiv 1 \pmod 7$, found by Bézout. Each term $r_i M_i M_i^{-1}$ is $1 \cdot r_i$ for its own modulus and 0 for the others, so

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 \equiv 23 \pmod{105},$$

and $23 = 7 \cdot 3 + 2 = 5 \cdot 4 + 3 = 3 \cdot 7 + 2$ checks all three conditions.

3 / UNITS SPLIT TOO / $\varphi(15) = \varphi(3)\varphi(5)$

Invertible numbers modulo 15 are the pairs of invertible numbers modulo 3 and 5

a	$(a \bmod 3,\ a \bmod 5)$	a	$(a \bmod 3,\ a \bmod 5)$
1	$(1, 1)$	8	$(2, 3)$
2	$(2, 2)$	11	$(2, 1)$
4	$(1, 4)$	13	$(1, 3)$
7	$(1, 2)$	14	$(2, 4)$

There are $2 \times 4 = 8$ pairs and 8 units. The equation $x^2 \equiv 1$ has 2 solutions modulo 3 and 2 modulo 5, so it has 4 solutions modulo 15, namely $1, 4, 11, 14$, and 8 modulo 105.

4 / READING 2023, 2024 AND 2015 / MODULO 1001

Since $1001 = 7 \cdot 11 \cdot 13$, a number is a triple of remainders, and blocks of three reveal it

n	$n \bmod 1001$	$\bmod 7$	$\bmod 11$	$\bmod 13$
2023	21	0	10	8
2024	22	1	0	9
2015	13	6	2	0

Because $1000 \equiv -1 \pmod{1001}$, the alternating sum of the three-digit blocks of 2023, which is $23 - 2 = 21$, is congruent to 2023 modulo 1001 and so shares its residues: $21 \equiv (0, 10, 8)$, which is why $7 \mid 2023$.

5 / WHEN THE MODULI SHARE A FACTOR

A system is solvable only if its conditions agree modulo the HCF

For $x \equiv 2 \pmod 4$ and $x \equiv 0 \pmod 6$ the HCF is 2 and both conditions say x is even, so the solutions form one class modulo the LCM: $x \equiv 6 \pmod{12}$. For $x \equiv 1 \pmod 4$ and $x \equiv 0 \pmod 6$ the first makes x odd and the second even, so there is no solution.

ASIDE / ARITHMETIC IN PIECES

To compute $2^{100} \bmod 15$ work modulo 3 and 5: $2^{100} \equiv 1$ and $\equiv 1$, so it is 1.

LINKS TO OTHER SHEETS

The inverses come from *Euclid's algorithm and Bézout's identity*; the test for $7, 11, 13$ is in *Primes, the sieve and divisibility tests to 13*.

Convention. $\mathbb{Z}/n$ is the integers modulo n; $\cong$ means a bijection that respects addition and multiplication.